

**INSIDE-OUT: Emotion Recognition through Handwriting Analysis using
Hybrid Approach**

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ABSTRACT

Emotion recognition from handwriting remains difficult due to the low accuracy and reliability of existing methods. These limitations reduce its usefulness in fields such as education, psychology, and forensics. This study introduces a hybrid approach that combines image processing techniques with machine learning to classify two emotional states, happiness and sadness, from handwriting samples. Data will be gathered from participants through self-reports and controlled induction methods, ensuring accurate emotion labeling. Each assessment will undergo preprocessing to extract key handwriting features, including slant, spacing, letter formation, and stroke dynamics. A hybrid architecture will then be employed, using Convolutional Neural Networks (CNNs) for feature extraction and Hidden Markov Models (HMMs) for temporal sequence classification. Inspired by frameworks such as EMOTHAW and TEemoDec, the system aims to achieve an accuracy rate of 80–85 percent. Beyond technical performance, the study highlights applications in education, where handwriting analysis could provide insights into students' emotional states and support timely intervention. Ethical issues such as privacy, bias, and responsible use are also considered. Overall, this research demonstrates the feasibility of accurate handwriting-based emotion recognition and extends its potential applications in both academic and professional domains.

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CHAPTER 1

INTRODUCTION

The first chapter of INSIDE-OUT: Emotion Recognition through Handwriting Analysis Using Image Processing Techniques presents the problems that motivated the conduct of this study. It also outlines the intention of addressing these issues through a systematic and technology-driven approach.

Emotions are complex psychological-physiological states that profoundly influence our thoughts, behaviors, and decisions" (Lerner et al., 2015). Emotions are subjective feeling states, characterized by a uniquely felt feeling component, such as happiness, sadness, fear, or anger, accompanied by a concomitant physiological response, such as changes in heart rate, respiration, hormone levels, and by an expressive response, such as facial expressions and body language. Essentially, emotions arise from the evaluation of events, situations, or stimuli as relevant to one's well-being or goals and serve important functions, such as ascribing meaning to experiences, facilitating social communication, and eliciting adaptive responses that aided survival across human evolution. Awareness and an understanding of emotions are crucial because they shape our perceptions, judgments, relationships, and goal-directed behaviors and thus influence nearly every aspect of our daily lives and decision-making processes.

Handwriting is the skill of developing letters, words, and sentences with the help of a hand, writing tool such as a pen or pencil, and a writing surface like paper (Deshpande, 2016). It is an extremely complex neuromuscular exercise coordinating the hand and eye, motor skills, and cognitive processes, and every person has his or her distinctive style of handwriting, which even shows character and emotional states and can be used as an identification tool (Durga & Deepu, 2018). It is also considered an art form with numerous calligraphic techniques (Samsuryadi et al., 2021).

Knowing and understanding handwriting are important because it will, by an individual's style of handwriting, provide clues to his or her personality and emotional well-being and indicate suitability for given roles or certain tasks and, thus, being a very useful tool in numerous fields, from psychology and forensics to human resources (Esposito et al., 2019).

1.1 Rationale of the Study

Understanding how emotions affect handwriting is really important. Emotions are key for thinking, deciding, and behaving. And how emotions show in handwriting can tell us a lot about someone's feelings. Looking at handwriting has been used in things like solving crimes, knowing someone's personality, and in psychology. But now, with better tech, we can learn even more about how emotions and handwriting are linked (Fairhurst et al., 2017).

If we can understand how emotions shape handwriting, we can make cool new tools. For instance, in school, looking at how kids write could help see if they're having a hard time or feeling stressed. Then, we could help them sooner (Robles-Bykbaev et al., 2021). In health care, seeing changes in handwriting could help spot brain or mood issues tied to how emotions are handled (Schröter et al., 2003). And in crime work and safety, checking handwriting for emotions could boost how we check docs and see lies (Gritčina et al., 2021).

Emotion are tied to thinking and actions, like handwriting. Handwriting, seen as just writing down words, shows feelings too. By studying how words are written like pressure, slant, spaces, and strokes, we learn about the writer's emotions (Fairhurst et al., 2017; Gritčina et al., 2021). This link between feelings and handwriting has use in many fields, like psychology, schools, crime, and health care.

In psychology, the link between feelings and handwriting helps make new tests and treatments. By looking at handwriting, docs may find emotional issues, like fear, sad days, or mood ills, and treat them better (Giezendanner

et al., 2018; Robles-Bykbaev et al., 2021). In schools, tracking changes in how kids write could show up early hints of stress, mood issues, or trouble learning, and help with care and plans (Guyon et al., 2019).

Also, the link between emotions and handwriting is big for crime work. Handwriting checks help see lies, make threat calls, and find who wrote things

(Gritčina et al., 2021). By finding true emotions in notes, crime pros can know the writer's mind, which may help spot who did bad things or why.

In healthcare, watching for shifts in handwriting could help find brain or mind ills tied to strong feelings, like Parkinson's, Alzheimer's, or schizophrenia (Schröter et al., 2003). Early finds could mean faster help and a good end.

To make most of the link between emotions and handwriting, we need help from lots of fields, like mind work, tech, math, and the study of lots of data forms. By using hard ways to find traits, deep ways to think, and ways to mix up lots of forms of data, we can make strong plans to look at the mix of emotions and handwriting (Fairhurst et al., 2017; Guyon et al., 2019). This mix of fields might lead to big finds, make new ways to think, and bring on neat plans that work in lots of spots, that help us know more of how folks act and feel.

The plan for this research is to link the fields of emotion psych and handwriting by using top tech like computer vision, deep learning, and ways to mix lots of data. By pulling in tricks like deep nets and hidden marks, we can show how emotional states and handwriting join up (Fairhurst et al., 2017; Guyon et al., 2019). This mix of skills could teach us new stuff, grow what we know, and make new tools that can help in school, health care, crime work, and more (Schröter et al., 2003; Gritčina et al., 2021).

1.2 Statement of the Problem

This study aims to develop and evaluate a system for emotion recognition from handwriting. This system will utilize advanced image processing and machine learning techniques.

1.2.1 General Objective

The general objective of this study is to develop and evaluate a machine learning system capable of classifying emotional states from student handwriting samples. Inspired by the EMOTHAW database, apply advanced image processing techniques.

1.2.2 Specific Objectives

To achieve this, the following specific objectives will be pursued:

1. Conduct data collection by gathering diverse handwriting samples from participants.
2. Preprocess the data.
3. Design and develop the CNN-HMM Model to be use to classify and identify the participants emotional state.
4. Test and determine the accuracy of the model obtained

1.3 Significance of the Study

The outcomes of this study may be relevant and instrumental to the following parties:

Human-Computer Interaction (HCI): By developing a novel emotion recognition system from handwriting, this research expands the ways humans can interact with technology. It offers the potential for more emotionally sensitive interfaces and adaptive systems that personalize experiences.

Emotional Intelligence: Understanding the link between handwriting and emotions helps unravel the complexities of human emotional expression.

This research could inform future development of tools that assist individuals in understanding their own emotions and potentially augment emotional self-regulation.

Psychology and Education: This work could provide valuable insights for psychologists seeking supplementary methods for assessing emotional states. Within education, emotion-aware technologies could help tailor learning experiences and create supportive environments.

Healthcare: While not a primary focus, this research could lay the groundwork for future exploration of handwriting-based emotion detection as a potential supplementary tool in mental health screening or monitoring.

1.4 Scope and Limitations

The scope of this study is to develop a handwriting-based emotion recognition system capable of distinguishing between two emotional states, happiness and sadness. To achieve this, handwriting samples will be collected from at least 300 right-handed participants between the ages of 18 and 25 at the University of San Carlos in Cebu City, Philippines. Each participant will complete a structured four-page assessment that includes instructions, a self-report questionnaire, drawing exercises, and writing exercises in both cursive and non-cursive styles. The responses will be written on short bond paper (letter size 8.5 × 11 inches) using a uniform 0.7 mm black pen provided by the researcher. All assessments will be digitized using an EPSON EcoTank L3210 scanner under consistent scanning conditions. From the digitized samples, timing-based features (such as stroke length and spacing patterns), ductus-based features (such as pressure dynamics and stroke curvature), and micro-level shape descriptors (including letter slant, alignment, and stroke-width variation) will be extracted. These features will be used as inputs to a hybrid architecture that combines a Convolutional Neural Network for spatial feature learning with a Hidden

Markov Model for temporal sequence classification. System performance will be measured through stratified 5-fold cross-validation, with evaluation metrics including macro-averaged F1-score and overall accuracy.

Despite this structured design, several limitations are acknowledged. First, the accuracy of the system depends on consistent scanning conditions and the handwriting styles of right-handed individuals within the 18–25 age range. The exclusion of left-handed participants and other age groups limits the generalizability of the findings. Second, emotional labeling relies on self-reports and controlled induction protocols, which may be influenced by social desirability or may not fully reflect spontaneous emotional states. Third, challenges such as smudged or poorly scanned samples, incomplete handwriting inputs, and the computational demands of running CNN–HMM models on low-resource devices are outside the scope of this research. Finally, cultural and demographic factors, such as differences in handwriting conventions and educational background, are not systematically considered. Ethical issues, particularly those related to data privacy, algorithmic bias, and responsible use in sensitive contexts like mental health, will require further validation and safeguards in future applications.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter discusses various studies related to INSIDE-OUT: Emotion Recognition through Handwriting Analysis Using Image Processing Techniques, with emphasis on emotion recognition through handwriting and image processing methods.

The ability to identify the emotional state, such as negative emotions like depression, anxiety, or stress, is a potential promise of handwriting analysis. Kedar (2015) and Kedar (2019) pinpoint the potential of handwriting analysis. LikformanSulem (2017) supports this by developing a database to recognize emotional states from handwriting and drawing. Ahamad (2023) extends this to the model of combining machine learning, deep learning, and natural language processing techniques to classify emotions from handwritten and text statements. This research collectively highlights the potential of handwriting analysis as a tool for emotion recognition.

Most of the previous work in emotion recognition through handwriting has been focused on deep learning techniques (Adebiyi, 2023). Most of the studies are aimed at sub solving subtle and complex emotion recognition, reducing the subjectivity and the inter-annotator variability, and increasing the accuracy. However, robust deep learning models, integration of multiple modalities, and the integration of physiological signals with behavioral signals are still required to fill this gap in current research as more sophisticated deep learning models are bound to evolve. Future research should consider the trend and development of emotion recognition using physiological signals (Li, 2015; Zhang, 2015) and speech (Cuihua, 2005) for developing comprehensive and accurate emotion recognition models.

Studies of emotion recognition through handwriting have addressed the motor theory, development, and learning from various psychologists, including Kao (1986). It was argued that handwriting can reflect emotional

stress, though this was hard to disentangle from acceleration as such. According to Kedar, in 2015, there was a technique for automatic emotion recognition through handwriting analysis, focusing on characteristics such as baseline, slant, pen-pressure, size, margin, and zone. Machine learning methods have been used in this area, but primarily within the context of computer recognition of handwriting (Plamondon, 1989). The impact of gender bias on emotion recognition through handwriting has not been directly addressed in the literature.

Emotion Recognition through Handwriting

Likforman-Sulem et al. (2017) developed the innovative EMOTHAW database, marking a significant step toward utilizing handwriting and drawing for emotional state recognition. Their research examined the correlations between handwriting features, such as timing and pen pressure, and measures of anxiety, depression, and stress. Using advanced machine learning techniques (random forests), they identified specific features indicative of particular emotional states, achieving promising results in classifying anxiety and stress. This study underscores the potential of handwriting analysis as a non-invasive tool for understanding emotional states, which could have applications in mental health screening and wellbeing promotion.

Kedar et al. (2015) provided a comprehensive review of automatic emotion recognition through handwriting analysis. Their work emphasized the potential connection between handwriting characteristics (such as baseline, slant, pen pressure, etc.) and an individual's emotional state. Importantly, they highlight the potential of handwriting analysis as a tool for identifying individuals who may be experiencing emotional distress and require psychological support. This review underscores the value of investigating the subtle cues embedded within handwriting, offering a

foundation for my own work on the relationship between handwriting patterns and stress levels.

Azmi, Fathima, and Wasid (2023) explored the feasibility of using handwriting analysis to detect emotions. Their study analyzed a dataset comprised of 129 users and examined the relationship between handwriting features and emotional states. Employing techniques like correlation analysis and dimensionality reduction, they developed a Random Forest model that demonstrated promising accuracy in classifying emotions. This research highlights the potential of handwriting analysis as a non-invasive tool for emotional detection, opening up possibilities for applications in mental health screening and support, a focus that parallels my own research on subtle behavioral cues of emotional states.

Gahmousse, Gattal, Djeddi, and Siddiqi (2020) explored the use of textural features in handwriting for personality identification. Building on the HWxPI Dataset and employing the Five Factor Model for personality, they applied Oriented Basic Image Features Column (oBIFC) and RankSVM to predict personality traits from handwriting samples. Their research found support for a relationship between personality traits and the visual characteristics of handwriting. This study highlights the potential for handwriting analysis in uncovering personality traits, which complements my research into the connection between nonverbal cues and psychological states.

Kedar and Kulkarni (2015) offer a comprehensive review of methodologies used in personality identification through handwriting analysis. They emphasize the potential of handwriting as a reflection of an individual's underlying personality traits. Their work highlights the significance of handwriting analysis in understanding how an individual's personality might influence their behavior and interactions. This study aligns

with my research interests in exploring non-verbal cues for personality assessment and the application of such techniques in realworld settings.

Bhattacharya, Islam, and Shahnawaz (2022) explored the use of Agglomerative Hierarchical Clustering to detect emotions from handwritten text samples. Their innovative approach aimed to bypass the limitations of traditional handwriting analysis, which can be time-consuming and dependent on expert interpretation. By clustering image pixels based on intra-cluster distances, they were able to classify emotions like anger, sadness, depression, happiness, and excitement with promising accuracy rates. This work suggests potential applications in mental health assessment, human resources, and even criminal investigations, emphasizing the importance of unsupervised learning techniques in emotion detection from handwriting.

Monica S, Kumar, Anaga, and Anagha (2020) explored the relationship between handwriting characteristics and mental health status. Their work suggests that handwriting analysis could potentially be used as a tool for identifying individuals who may be experiencing negative emotional states like depression, anxiety, and stress. This study highlights the importance of considering nonverbal cues as potential indicators of mental health and aligns with my own research exploring the connections between emotional well-being and subtle behavioral markers.

Handwriting Analysis

Handwriting analysis is one such complex field that focuses on the study of individual expression through manual, aesthetic, emotional, and intellectual characteristics (Paget, 1940). Recent developments have witnessed the formulation of a dynamic model for handwriting, wherein a differential equation is used to model time-varying behavior. For example, Yizhi developed a differential equation to model time-varying behavior.

Online fuzzy models were applied by Bouillon (2015) in the analysis of handwriting quality, with an aim towards teaching children how to write. The practical application of handwriting analysis was entailed by Bensefia (2004) wherein segmentation procedures and statistical tests were incorporated in order to verify the writer. All these studies, together, give an insight into the multi-dimensionality of handwriting analysis with its corresponding applications.

Samsuryadi, Kurniawan, and Mohamad (2021) conducted a comprehensive survey on automated handwriting analysis within the framework of pattern recognition. Their work highlights the broad applications of handwriting analysis, including its uses in recruitment, medical diagnosis, and psychology. They emphasize the potential of computerized analysis to support the work of graphologists in identifying personality traits. Importantly, their survey details the various stages of the pattern recognition process, focusing on data collection, feature extraction, and classification techniques. This study provides a valuable overview of methods and advancements in the field of automated handwriting analysis.

Akhtar et al. (2021) explored the use of pen-tablet technology to analyze handwriting for both person identification and handedness determination. Their study focused on extracting distinguishable features from handwriting samples and applying classification algorithms (Support Vector Machines and Random Forest Classifiers) to achieve identification. This work aligns with the increasing interest in automated handwriting analysis and its potential to offer insights beyond traditional manual methods, a connection which is relevant to my own investigation into handwriting and personality traits.

Esposito et al. (2019) explored the relationship between quantitative handwriting and drawing features and personality traits as measured by the Big Five Personality questionnaire (BFQ). Analyzing data from 78

participants, they found significant differences in features like pressure, speed, and spacing between individuals scoring low, typical, and high on the BFQ dimensions (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism). Their study highlights the potential connection between graphology and the measurement of personality traits, suggesting that handwriting analysis may provide insights into an individual's personality.

Sesa-Nogueras (2011) delved into the potential of online handwritten words for writer recognition purposes. While much of the existing research in this area has concentrated on signatures, this study turned its attention to everyday handwriting. Employing a unique recognition system utilizing stroke categorization and dynamic time warping methods, the findings revealed that short online text sequences (individual words or word combinations) exhibited remarkable effectiveness in both writer identification and verification tasks. Using a dataset encompassing 320 writers and multiple repetitions of Spanish words, single-word identification rates spanned from 80.6% to 95.6%, with verification error rates between 1.57% and 5.55%. When combining two words, the top identification rate soared to 99.7%, and the best verification error rate dropped to 0.63%.

Machine Learning and Image Processing Techniques

Several studies were conducted to explore the combination of machine learning and image processing techniques. Millán gave evidence in 2018 that the best method for image classification, especially in a complex set of data, was the Transfer Learning/Image Retraining technique. Rocha, in 2009, suggested using classifiers and machine learning techniques for image categorization, authentication, and multi-class classification. Content is evaluated by Thakur in 2019 through machine learning, specifically in the image retrieval methods. In 2016, Adithya proved that machine learning can be put in the application of many fields such as science, technology, and agriculture through a comprehensive survey of machine learning usage in

image processing techniques. All the above studies combined reflect the potential of using machine learning and image processing techniques to improve image recognition and processing.

Nadeem, Ahmad, and Habib (2022) explored the detection of emotional states from handwriting using machine learning techniques. Their research emphasizes the connection between handwriting characteristics and underlying emotions, positing that handwriting patterns can indicate emotional states. The study aimed to broaden the understanding of emotion detection from handwriting, aiming to support interventions for mental health by identifying individuals experiencing emotional distress. Utilizing machine learning models, this work contributes to the growing body of research on emotion recognition using nontraditional modalities.

Joshi, Ghaskadbi, and Tendulkar (2018) explored the use of machine learning techniques for employability evaluation through handwriting analysis. Their research draws upon the principles of graphology, suggesting a link between handwriting features and a person's psychological state. The authors propose methods using image processing techniques to extract relevant handwriting features from samples. These features were then analyzed using classifiers to assess a candidate's employability from an HR standpoint. This study highlights the potential of handwriting analysis as a tool within the recruitment process.

One relevant study the impact of pre- and post-image processing techniques on deep learning frameworks in digital pathology image analysis is Salvi et al. (2021). In their comprehensive review published in *Computers in Biology and Medicine*, the authors provide an overview of the methods used for preprocessing inputs and postprocessing outputs within deep learning pipelines for digital pathology image analysis tasks like classification, detection, and segmentation. They highlight how integrating traditional image processing methods as pre- and post-processing steps can further enhance

the performance of deep neural networks compared to using the networks alone. The study underscores the importance of carefully considering image preprocessing and postprocessing techniques when designing deep learning frameworks for effective digital pathology image analysis.

Convolutional Neural Networks and Hidden Markov Model

Many studies investigated the combination of Convolutional Neural Networks and Hidden Markov Models across diversified applications. Regli, 2016, introduced a scattering convolutional hidden Markov tree that has witnessed promising results in image classification. Sarmiento, 2020, compared the performance of HMMs and CNNs in image classification and revealed that HMMs can achieve competitive results with less computational resources. Baucum, 2020, proposed hidden Markov recurrent neural networks, which are a special case of recurrent neural networks and show that its likelihood function is the same as that of discrete-observation HMMs. The models were used for disease progression modeling, and the parameter estimates fit well to the patient data than traditional HMMs. Lin, in 2019, introduced a neural hidden Markov model, in which a neural network computes transition probability in higher-order HMMs, so that the performance in time-aware recommender systems has improved. All together, these studies reflect that the combination of CNNs and HMMs is very promising in various applications, such as image classification, disease progression modeling, and recommender systems.

Kulakov (2021) and Udeh (2021) both explore the use of convolutional neural networks (CNNs) for emotion recognition, with Kulakov focusing on keyboard handwriting and Udeh on facial expressions. Le (2013) introduces a hybrid model of Hidden Markov Models (HMMs) and Deep Belief Networks for emotion recognition from spontaneous speech, achieving state-of-the-art results. Bharti (2022) proposes a hybrid model for text-based emotion recognition, combining CNNs and Bi-GRU with a support vector machine.

These studies collectively demonstrate the potential of hybrid models, particularly those incorporating CNNs, for emotion recognition across different modalities.

Goumiri, Benboudjema, and Pieczynski (2023) propose a hybrid classification model (CNN-HMC) that combines convolutional neural networks (CNNs) for feature extraction with hidden Markov chains (HMCs) for the classification step. The Peano scan is used to create multiple HMCs, and the Expectation-Maximization (EM) algorithm enables unsupervised classification using the Bayesian Maximum Posterior Mode (MPM) method. Their model aims to improve classification accuracy compared to traditional CNNs. Experiments demonstrate CNN-HMC's superiority over standalone CNNs (like 4Conv and Mini AlexNet) as well as various hybrid models using SVMs, LSTMs, Random Forests, and gcForests as classifiers.

Tso and Olsen (2005) explored ways to improve the accuracy of unsupervised image classification in remote sensing. Traditional k-means clustering, relying solely on spectral data, often produces suboptimal results and visually "busy" images. They proposed using Hidden Markov Models (HMMs) to incorporate both spectral and spatial information, leading to smoother output and enhanced classification accuracy. Their methodology adjusted observation sequences and observation densities, integrating 2D spatial data into the HMM. Results demonstrated that combining spectral and spatial data within the HMM framework yielded superior classification accuracy and higher quality visuals compared to standalone k-means clustering.

CHAPTER 3

TECHNICAL BACKGROUND

This chapter provides a comprehensive discussion of the theorems, definitions, fundamental algorithms, mathematical models, and formulas relevant to INSIDE-OUT: Emotion Recognition through Handwriting Analysis Using Image Processing Techniques.

Emotion Recognition

Emotion recognition is the identification and interpretation of human emotions that come with a lot of factors. Wagle, in 2020, points out that technology has been instrumental in improving this, especially by collecting and analyzing the available data. Frijda, in 1969, states that knowledge and experience play an important role in this process and that it involves the use of conscious hypotheses and self-corrections. Hurst, in 2012, puts it that emotion is subjective, and any technological advancements in this area promise to make the recognition of any emotion a bit easier.

Hand analysis

Hand analysis, otherwise called graphology, is a way of identifying and understanding personality traits through handwriting. In such a method, a range of handwriting features such as the size and location of page margins, spacing between the lines and between words, the slope of words, and speed of writing are analyzed and determine the personality of the handwriter. The results of such analysis can be automated in the use of image processing techniques where the human hand is minimized. There is uniqueness in one's handwriting that mirrors the character and the mood of a person. This graphology is used in writing in scripts, such as Farsi, Urdu, and Arabic (Khan, 2019).

Data

Data is a collection of facts obtained from an object or a group of objects with the purpose of being studied to create something meaningful and useful ("Data", 2021). A database is a collection of data, organized in a systematic manner for the ease of retrieval, management, and manipulation ("Database Defined", 2021).

Data Set

A Dataset is a set or collection of data. This set is normally presented in a tabular pattern. Every column describes a particular variable. And each row corresponds to a given member of the data set, as per the given question (Byjus, 2024).

Image Processing

Image processing involves the transformation of input images into output images or image characteristics (Katariya, 2018). This process can enhance images for human viewing or interpretation and is used in various fields such as optical imaging, medical imaging, and computer vision (Lowe, 2015). It is a computational process that can improve image quality and extract features for further analysis (Corke, 2017). Digital image processing specifically involves the manipulation of pixel gray levels and their distribution to produce a better-suited output image (Trivedi, 2018).

Machine Learning

Machine learning is a branch of computer science that uses existing data to predict future data (Paluszek, 2017). This involves mathematical optimization, statistics, and algorithm design (Arora, 2019). A very prominent category of artificial intelligence makes use of data to train the computer to perform some tasks (Schneider, 2018). A great challenge in machine learning

is that models often need to be dependable and explainable, and this can be achieved through automated reasoning (Bride, 2018).

Machine Learning Model

A model is a physical representation of abstract concepts or is an imitative representation of another object (Merriam-Webster, 2024). A machine learning model is a program that can find patterns or make decisions from a previously unseen dataset. For example, in natural language processing, machine learning models can parse and correctly recognize the intent behind previously unheard sentences or combinations of words (Databricks, 2024).

Convolutional Neural Network

A Convolutional Neural Network (CNN) is a type of deep learning algorithm specifically designed for image processing and recognition tasks. Compared to alternative classification models, CNNs require less preprocessing as they can automatically learn hierarchical feature representations from raw input images.

Hidden Markov Model

The Hidden Markov Model (HMM) is the relationship between the hidden states and the observations using two sets of probabilities: the transition probabilities and the emission probabilities. The transition probabilities describe the probability of transitioning from one hidden state to another (Geeksforgeeks, 2024).

Image Processing Techniques for Handwriting Analysis:

Image processing really helps in extracting and analyzing relevant features of the handwritten samples. Mishra 2020 and Tilagul 2022 highlight that OCR is used to identify and translate handwritten text into digital form,

thus bridging the gap between physical and digital data so that data manipulation and analysis can be done swiftly. Sen 2018 and Jubair 2013 show the use of image processing to extract some features of handwriting, which include the size and slant of the text and the frequency of predefined shapes of characters. These features could be used for analysis of personality and for character identification.

Machine Learning Algorithms for Emotion Recognition:

Emotion recognition using machine learning has been done in a variety of algorithms, but focusing on facial expressions in particular. Sharma (2019) compared K-nearest neighbors, Support Vector Machine, and neural network algorithms to give an insight into the performances of these algorithms for the same task. Zhang (2020) highlighted that multi-modal data, mainly EEG signals, could be even more objective and reliable for emotion recognition. Moreover, he reviewed the detailed methods of feature extraction and reduction and the design of ML classifiers. Pulido-Castro (2021) made an improvement in facial emotion recognition by developing an ensemble machine learning model that enhanced the recognition accuracy and reduced the execution time for the realization of real-time recognition.

Integration of Handwriting Analysis and Image Processing:

The synthesis of handwriting analysis and image processing involves the development of a cohesive framework. This framework encompasses the preprocessing of handwritten samples, feature extraction through image processing, and the utilization of machine learning algorithms for emotion classification. Many studies have integrated handwriting analysis and image processing. For example, Mishra 2020 used OCR and neural networks to detect and convert handwritten text into digital text. Such applications are useful in detecting forgery and assessing personality traits. Sen 2017 proposed an automated handwriting analysis system using graphology

principles with image processing for feature extraction. Premananda 2020 made a proposal for automated image handwriting recognition using Probabilistic Patch-based filters and Canny edge recognition. Mantri 2022 concentrated on handwritten digit recognition using image processing and machine learning to automatically recognize and exhibit digits in input images. All of these studies together suggest the potential of integrating handwriting analysis and image processing for such applications

CHAPTER 4

DESIGN AND METHODOLOGY

4.1 Research Environment and Respondents

The study entitled INSIDE-OUT: Emotion Recognition through Handwriting Analysis Using Image Processing Techniques will be conducted at the University of San Carlos in Cebu City, Philippines. The university environment provides a controlled setting that ensures both accessibility to participants and standardization of research conditions. Research data collection will be carried out by visiting classrooms before or after scheduled classes, allowing the researcher to invite students directly and explain the study's purpose. A minimum of 300 respondents will be recruited from the general student population through this classroom-based approach, as well as through posted announcements and coordination with faculty. The target group will consist of right-handed young adults aged 18 to 25 years. To broaden the range of handwriting features represented in the dataset, both cursive and non-cursive writing styles will be collected from each participant.

4.2 Research Instrument or Sources of Data

The primary research instrument will be a structured four-page assessment handout. It will be prepared and distributed by the researcher and will include the following sections:

1. **Page 1 – Instructions:** Clear guidelines for completing the assessment.
2. **Page 2 – Self-Report Assessment:** A questionnaire where participants indicate their emotional state.
3. **Page 3 – Drawing Exercises:** Tasks designed to capture graphical features of handwriting.
4. **Page 4 – Writing Exercises:** Both cursive and non-cursive writing samples for analysis.

Participants will complete the assessment on short bond paper (letter size 8.5 × 11 inches) using a 0.7 mm black pen. All required materials, including the pens and paper, will be provided by the researcher, and no expenses will be incurred by the participants. Completed assessments will be digitized using the EPSON EcoTank L3210 scanner to ensure uniform image quality for processing and analysis. To protect participant confidentiality, names will not be collected. Instead, each assessment will be assigned a unique barcode generated by the researcher, which will serve as the identifier during data management and analysis.

4.3 Research Procedure

4.3.1 Gathering of Data

The data collection for the study entitled INSIDE-OUT: Emotion Recognition through Handwriting Analysis Using Image Processing Techniques will be conducted at the University of San Carlos, located in Cebu City, Philippines. The university setting provides a controlled environment suitable for research, while also ensuring accessibility to the target population. Participants will be recruited from the general student population through direct invitations, posted announcements, and coordination with faculty members. The study aims to recruit a minimum of 300 right-handed young adults between the ages of 18 and 25, representing both cursive and non-cursive handwriting styles.

Each participant will be asked to complete a structured four-page assessment handout prepared by the researcher. The assessment includes instructions (page 1), a self-report questionnaire to capture the participant's emotional state (page 2), drawing exercises (page 3), and writing exercises in both cursive and non-cursive styles (page 4). All responses will be written on short bond paper (letter size 8.5 × 11 inches) using a 0.7 mm black pen. Participants will not incur any expenses for the study, as all materials, including the pens and paper, will be provided by the researcher. After completion, the assessments will be collected and

scanned using the EPSON EcoTank L3210 to produce high-quality digital copies. To maintain confidentiality, participants will not provide their names; instead, each assessment will be labeled with a unique barcode assigned by the researcher for secure identification during analysis.

4.3.2 Treatment of Data

The scanned handwriting assessments will undergo several preprocessing steps to ensure consistency and quality before feature extraction. These steps include binarization to separate text from the background, noise removal to eliminate scanning artifacts, skew correction to align the writing properly, and normalization to standardize image dimensions. Following preprocessing, relevant handwriting features will be extracted, focusing on aspects such as letter slant, spacing, stroke dynamics, pen pressure, and shape consistency.

Each dataset will then be matched with emotional labels derived from the participants' self-report questionnaires and controlled emotion induction protocols. These labeled datasets will be organized systematically and prepared as input for training and evaluating the hybrid model. The Convolutional Neural Network (CNN) will be used to capture spatial handwriting features, while the Hidden Markov Model (HMM) will analyze temporal patterns. Together, these models will classify the handwriting samples into the emotional states of happiness and sadness. This process ensures that the data is both reliable and valid for use in achieving the objectives of the study.

4.4 Conceptual Framework

This framework serves as an initial evaluation of recognizing emotional states from handwriting data using Convolutional Neural Networks (CNN) and Hidden Markov Models (HMM), along with image processing techniques.

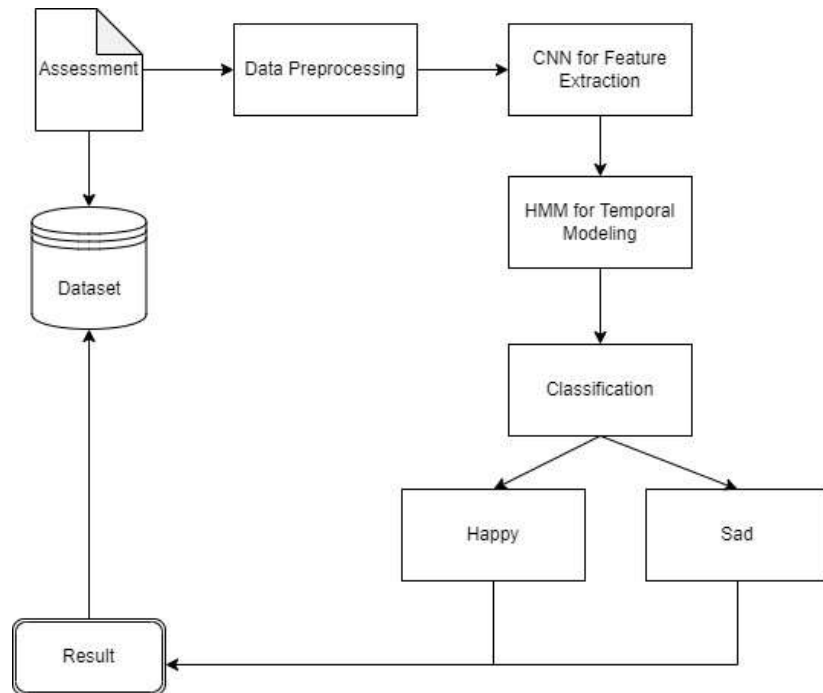


Figure 1: Conceptual Framework

Figure 1 illustrates the overall flow of the proposed methodology for emotion recognition through handwriting analysis. Preliminary assessment of the user is taken as the first step, and the subsequent steps are accordingly progressed. The phase of data collection gathers handwriting samples from participants and ensures that the dataset is different but representative. The cleaned, normalized, and preprocessed data collected from the handwriting samples are then fed into other data cleaning techniques. The next step is the extraction of relevant features, whereby the extraction techniques can be image processing methods. The development of the model follows, whereby CNNs are designed for feature extraction and developing the hidden Markov models for emotion classification. The integrated CNN-HMM model is then evaluated through cross-validation and other performance metrics for the classification done by the model to reflect its accuracy and effectiveness. The results and analysis phase include interpreting the model's performance in drawing conclusions and exploring its possible applications or areas for further improvement.

4.5 Analysis and Design

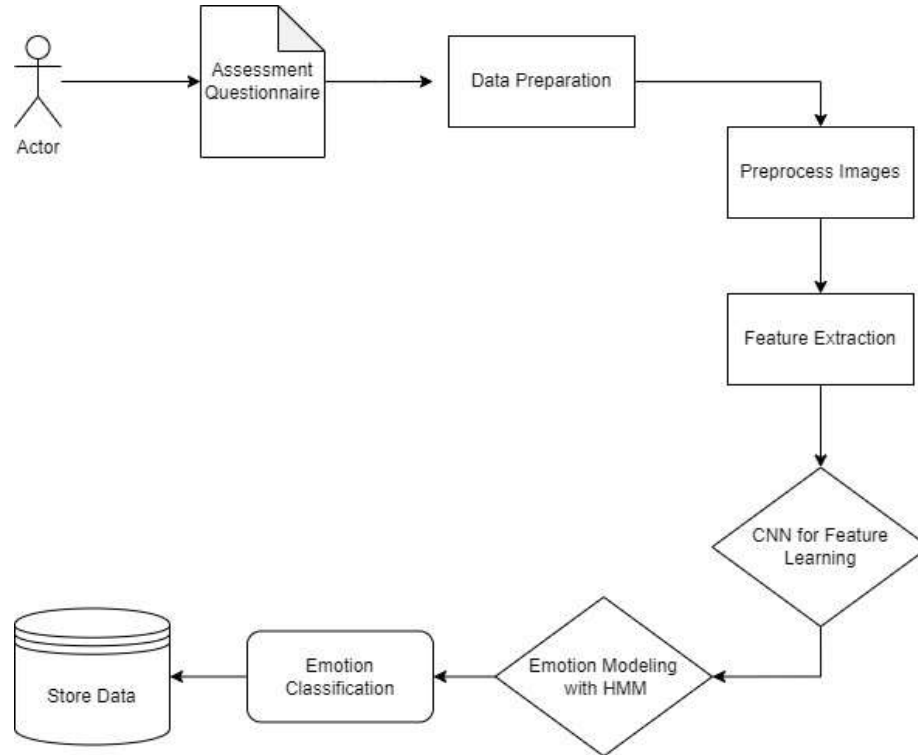


Figure 2: Data Acquisition Flow

To automatically identify emotions from a preprocessed image from the assessment. The process begins with an evaluation followed by the preparation of data. The visual input is fed into a convolutional neural network to get the features. A dynamic model, by a Hidden Markov model, is used to represent the emotions as time progresses. The characteristics inferred and the dynamic modeling are later combined to classify into two emotional states: happiness and sadness. The end product is a recognizing and classifying system of emotions. The system has the potential for application in human-computer interaction, affective computing, and emotional intelligence analysis.

4.6 Development Model

As a solo researcher, I chose to use the Agile Framework known as Kanban, known for its flexibility and ability to transform organizations at my own pace ([Shalloway, 2012](#)). It is particularly effective in continuous-flow environments, emphasizing system improvement ([Zayat, 2020](#)). In the classroom, it has been found to promote self-organizing teams and regular reflection, leading to improved communication and coordination ([Saltz, 2020](#)). In software development, Kanban provides visibility into work in process, resource capacity, and workflow, making it an effective and low-cost method ([Reese, 2011](#)).

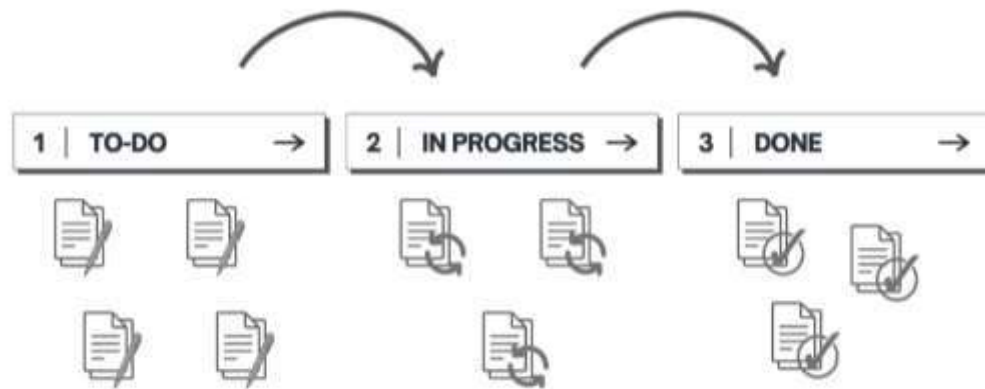


Figure 3: Visual Representation of Kanban Framework

Kanban boards are a popular tool for project management, with a focus on task tracking and status updates ([Bacea, 2017](#)). They are particularly useful for breaking down projects into individual tasks and visualizing progress, with a typical classification into "to-do", "doing", and "done" ([Bieniusa, 2018](#)).

As a solo researcher, the Kanban board is great in that it gives a tangible view of what needs to be done in each stage of a project in a structured and efficient order. Typically, there is a core of three columns in the traditional board: "To Do," "In Progress," and "Done." Being a flexible system, it allows the inclusion of columns specifically designed to fit the needs and complexities of the research process.

The "To Do" column is a single repository of all the tasks and requirements that should be addressed. It allows the researcher to plan and divide large tasks into smaller and more manageable units and set priorities. An example of items in this column for a thesis proposal might be "literature review," "methodology," "data collection planning," and "identifying software tools".

Tasks move from the planning column to the "In Progress" column as work on them is actively begun. The tasks in this column are very clearly what is being worked on, which further helps the researcher focus and track efficiently. Suggestions for revision from advisors or reviewers would normally stay in this column until fully addressed.

The "Done" column makes one feel accomplished and displays some amount of progress in completing tasks; in the case of a thesis proposal, this could be the milestones, from the ideation of the topic to its definition, planning, and initial development stages, which give way to the accomplishment already gained.

For a solo researcher-visualizing through a Kanban board-tracking the progress of the project and any bottlenecks would be easily kept. This will help them make decisions throughout the lifecycle of the research, enhancing productivity and ensuring a structured approach to the project.

4.7 Development Approaches

The development of INSIDE-OUT utilizes the bottom-up development approach. This means that the study and the system come as a result of progress made by using and combining different elements.

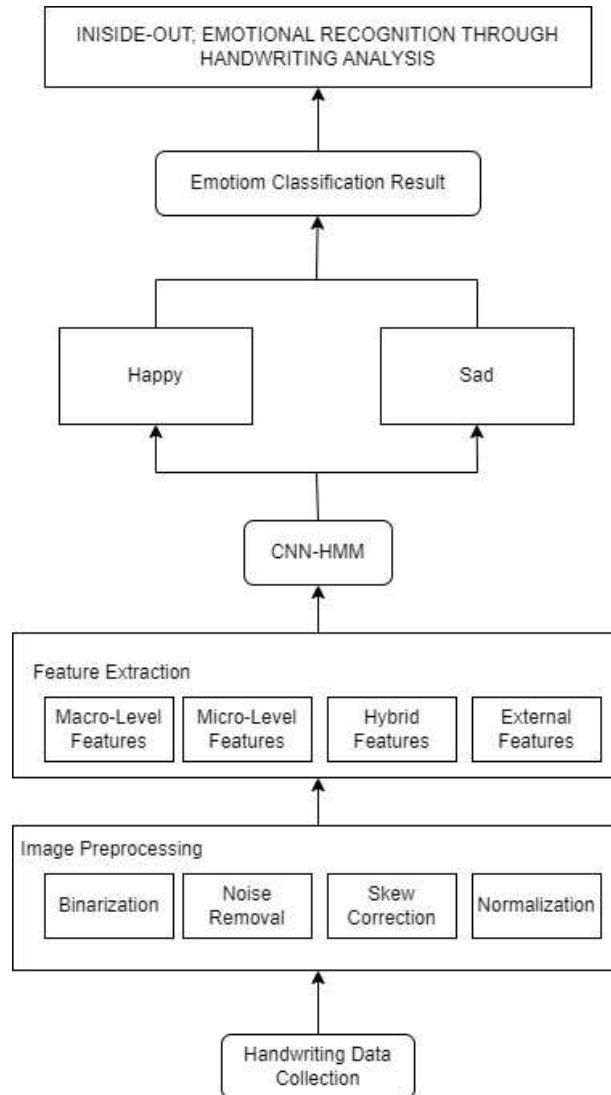


Figure 4: Bottom-Up approach for INSIDE-OUT

Figure 4 follows a data-driven approach, collecting a data set of handwriting samples with annotations conducted by both self-reported and controlled experiments of inducing emotions. This raw data is preprocessed using various image preprocessing techniques like binarization, noise removal, skew correction, and normalization to get ready for feature extraction. Macro-level features, namely, slant, spacing, micro-level features, namely, letterforms, strokes, hybrid features, namely, macro and micro features, and self-reports or the controlled experiments of emotion induction are extracted from preprocessed data.

Then these features are fed into a CNN-HMM model for training and testing to understand the patterns and links between handwriting characteristics and emotional states. The trained CNN-HMM model can classify the handwriting samples in two categories of emotions, namely happiness, and sadness, to give the final output of emotion classification.

4.8 Software Development Tools

There are numerous tools, software and library packages used in its execution and development of the intended system of this study. The software used with their appropriate versions, as well as their use, are enumerated in Table 1 below.

Software	Version	Use
Google Chrome	124.0.6367.60	A versatile web browser used for accessing websites, online applications, and various internet-based services.
Visual Studio Code	1.87.2	A powerful code editor for building and debugging applications across various programming languages.
GitHub	3.12.2	A platform for collaborating on software projects, version control, and code sharing.
TensorFlow	19.0	A comprehensive library facilitating the development and training of machine learning and deep learning models.
OpenCV	4.9.0	A library focused on real-time computer vision applications such as image processing, video analysis, and object recognition.

PyCharm	2024.1	An integrated development environment (IDE) specifically designed for efficient Python programming.
Scikit-learn Package	1.4.2	A library providing a range of tools for machine learning tasks, including classification, regression, clustering, and data analysis.
Draw.io	24.2.5	a web-based diagramming tool that provides a range of features, including flowcharting, network diagrams, and UML diagrams, among others

Table 1: Software Development Tools

4.9 Project Management

This section discusses the calendar and timeline, roles and responsibilities, and budget and cost management of the team at the time of the development of the research project. Each subsection outlines its scope and also explains the rationale for the decision made in the management of the project.

4.9.1. Schedule and Timeline

This subsection under project management indicates the timeline observed and followed by the researchers in compliance with submission deadlines as well as incremental research deliverables. Table 3 shows the list of tasks, scheduled according to the different stages in accomplishing the research. Table 2 shows the list of activities and deliverables to be accomplished within the 2nd semester of the Academic Year 2024 - 2025, with the objective of a thesis proposal.

Details			Q1								Q2							
Tasks		M	FEB				MAR				APR				MAY			
		W	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
1	Research Topic Ideation	Research Title and Theme																
2	Thesis Definition and Planning	CHAPTER 2 Review of Related Literature																
		CHAPTER 3 Technical Background																
		Statement of the Problem																
		CHAPTER 4 Design and Methodology																
3	Thesis Development	CHAPTER 1 Rationale of the Study, Significance of the Study, Scope & Limitation, and Abstract																
		Transmittal Letter, Bibliography, Assessment, and Curriculum Vitae																
4	Thesis Revision	Submission of Draft Thesis Paper																
		Initial Thesis Revision																
5	Thesis Defense	Submission of Thesis Paper																
		Thesis Proposal Defense																
		Thesis Paper Revision (Panelist Feedback)																
		Submission of Final Revised Thesis Paper																

Table 2: Gantt Chart of Activities, Second Semester, A.Y. 2024 – 2025

Table 3 shows the flow of work that will be observed during the second half of the completion of the thesis through the table of schedule. In the second

half, the actual employment of the proposed system/algorithm begins: this entails the training of the algorithm as well as the testing and verification, and validation of the accuracy of the obtained results.

Details																														
TASKS			JUN				JUL				AUG				SEP				OCT				NOV				DEC			
			1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
1	Data Gathering	-Collecting Data Handwriting Samples																												
2	Data Processing	-Preprocesses data and Feature Extraction																												
3	Data Training	-Training the Algorithm using the obtained data samples																												
4	System Testing and Verification	-Model Accuracy Testing and Reinforcement																												
5	Publish Paper	-Present Paper																												
6	Thesis Documentation	-Chapter 5: Results and Discussion -Chapter 6: Conclusion																												
7	Thesis Defense	-Document Revisions -Submission of Document and Thesis Defense Preparation -Thesis Defense -Thesis Paper Revision (Panelist Feedback) -Submission of Final Revised Thesis Paper																												

Table 3: Gantt Chart of Activities, Summer Term, A.Y. 2025, and First Semester, A.Y. 2025 – 2026

4.9.2. Budget and Cost Management

To estimate the total cost of this research, a breakdown of expenses is provided in Table 4. This table includes all items and equipment used throughout the study, along with their corresponding prices. This information can be a valuable reference for future research on this topic, allowing researchers to get a sense of the approximate costs involved.

Items	Cost
Laptop (LENOVO ideapad 310S)	42,000.00
Printer and Scanner (EPSON EcoTank L310)	10,000
Pens and Paper for Assessments	5,000
Conference Fees (includes the Conference Registration Fee, Airfare, Accommodation, etc.)	30,000.00
Miscellaneous	10,00.00
Total	97,000.00

Table 4: Table of Expenses

4.10 Verification, Validation, and Testing

Once the main objective of the study is accomplished, the quality of the study is then verified and validated. This includes the accuracy of the results based on the training and testing of the data set, along with the means by which the training and testing are done. The expected accuracy of the proposed handwriting analysis system for emotion recognition is at least 80 - 85%.

To ensure accuracy, reliability, and applicability of the proposed handwriting analysis system for emotion recognition. The validation process will include a vigorous analysis of system performance using benchmarks and metrics that are acknowledged in the light of previous work, for example, EMOTHAW, TEemoDec, and so on. The accuracy of the emotion classification

of the system will be tested on a large dataset of handwriting samples from different demographics, handwriting styles, and emotional states.

During the development and testing periods, a professional psychologist will be involved in the validation of the psychological underpinnings and real-world applicability of the system. He/she will review the emotion induction techniques used in the dataset to ensure validity and adherence to ethical standards. He/she will assess the appropriateness of self-report measures used for emotional annotations, considering potential biases or limitations. He/she will assess the performance of the system in recognizing emotions accurately, providing insights into the alignment of the system with established psychological theories and models of emotion. Their expertise is instrumental in interpreting the results and identifying areas for further work, exploring practical applications in settings such as education or other relevant domains.

Stratified k-fold cross-validation will be used to evaluate the performance of the hybrid CNN-HMM algorithm. The dataset is here split into k folds, and the algorithm is trained and tested on different folds iteratively.

The evaluation technique is robust, and the algorithm can prevent overfitting and make testing on unseen data possible. In the handwriting dataset, it is first stratified based on the emotion classes, hence making every fold contain a representative distribution of the classes. The dataset has been split into k equally sized folds or subsets. Then, the algorithm is trained on k-1 folds and tested on the remaining fold. This process is repeated k times, and every fold has once been used as the test set. A performance metric is computed for each fold and averaged to obtain an overall estimate of the algorithm's performance.

The most common evaluation metrics for multi-class classification tasks are the F1-score, which combines precision and recall into a single measure. F1-scores for each emotion class can be calculated using the following formula:

$$\mathbf{F1\text{-}score = (2 \times Precision \times Recall) / (Precision + Recall)}$$

$$\mathbf{Where: Precision = TP / (TP + FP) \quad Recall = TP / (TP + FN)}$$

Figure 5: F1 – Score Evaluation Metric

TP (True Positives) is the number of instances correctly classified as belonging to a particular emotion class. FP (False Positives) is the number of instances incorrectly classified as belonging to that emotion class. FN (False Negatives) is the number of instances of that emotion class that were misclassified.

In order to obtain an overall F1-score for the multi-class emotion recognition task, F1-scores for individual classes can be combined using either a weighted average or macro-averaging approach. Macro-averaged F1-score treats every class equally and can be calculated as follows:

$$\mathbf{Macro\text{-}averaged \ F1\text{-}score = (F1\text{-}score_class1 + F1\text{-}score_class2 + ... + F1\text{-}score_classN) / N}$$

Figure 6: Macro-averaged F1-score

Where N is the total number of emotion classes.

Another useful metric is the classification accuracy, which calculates the fraction of correctly classified instances out of the total number of instances.

Accuracy can be obtained as follows:

$$\mathbf{Accuracy = (TP + TN) / (TP + FP + FN + TN)}$$

Figure 7: Classification Accuracy

Where TN (True Negatives) is the number of instances correctly classified as not belonging to a particular emotion class.

Stratified k-fold cross-validation and such evaluation metrics will help in the assessment of the hybrid CNN-HMM model and provide a reliable and statistically sound estimate of its ability to recognize emotions in the handwriting dataset.

4.11 Ethical Considerations and Data Management Plan

In conducting this research, we will rigorously minimize any risk of harm by fully informing participants of the study's nature and voluntary procedures before data collection begins. Consistent with best practices in interview- and survey-based protocols, participants will be assured that no question is mandatory and that they may pause, reschedule, or withdraw at any point without penalty or loss of benefits. Informed consent will be obtained via a two-part form: an information sheet outlining the study's purpose, duration, and procedures, and a consent certificate confirming voluntary participation and permission to record responses. For our handwriting tasks, consent forms and questionnaires will clearly state the right to refuse or skip any item, and a brief debriefing will be offered to ensure participants leave without distress. Contact details for the researcher, faculty adviser, and the University of San Carlos Research Ethics Committee will be provided for any queries or in the unlikely event of research-related injury.

To safeguard confidentiality and data integrity, each participant will be assigned a unique code number; no names will appear on study materials. All physical documents (consent forms, questionnaires) will be locked in a cabinet, while digital assets (scanned handwriting images, extracted features, model outputs) will reside on a password-protected computer dedicated solely to this project. Access to these files is restricted to the researcher and adviser, both of whom will sign nondisclosure statements before being granted entry. A transfer logbook will record any data movements or sharing events, and a separate "incidental findings" folder will isolate unforeseen data for review. Finally, all electronic data will be hard-deleted and all paper records shredded two years after study completion to ensure irreversible disposal.

BIBLIOGRAPHY

- Esposito et al., "Handwriting and Drawing Features for Detecting Personality Traits," 2019 10th IEEE International Conference on Cognitive Infocommunications (CogInfoCom), Naples, Italy, 2019, pp. 79-84, doi: 10.1109/CogInfoCom47531.2019.9089985. keywords: {Task analysis;
- Analysis of variance; Writing; Feature extraction; Sea measurements; Tools; Conferences},
- Gahmousse, A. Gattal, C. Djeddi and I. Siddiqi, "Handwriting based Personality Identification using Textural Features," 2020 International Conference on Data Analytics for Business and Industry: Way Towards a Sustainable Economy (ICDABI), Sakheer, Bahrain, 2020, pp. 1-6, doi: 10.1109/ICDABI51230.2020.9325664. keywords: {Feature extraction;Writing;Training;Visualization;Data mining;Task analysis;Support vector machines;Personality Identification;Handwriting Analysis;oBIF Columns;Five Factor Model;RankSVM},
- Adebiyi, M.O., Fatinikun-Olaniyan, D., Adebiyi, A.A., & Okunola, A.A. (2023).
- Survey on Current Trend in Emotion Recognition Techniques Using Deep Learning. 2023 International Conference on Science, Engineering and Business for Sustainable Development Goals (SEB-SDG), 1, 1-9.
- Adithya, A., & KavyaRamesh (2016). MACHINE LEARNING IN IMAGE PROCESSING – A SURVEY.
- Arora, S. (2019). MATHEMATICS OF MACHINE LEARNING: AN INTRODUCTION. Proceedings of the International Congress of Mathematicians (ICM 2018).
- Baucum, M., Khojandi, A., & Papamarkou, T. (2020). Hidden Markov models are recurrent neural networks: A disease progression modeling application. ArXiv, abs/2006.03151.

- Bensefia, A., Paquet, T., & Heutte, L. (2004). Handwriting analysis for writer verification. Ninth International Workshop on Frontiers in Handwriting Recognition, 196-201.
- Bharti, S.K., Varadhaganapathy, S., Gupta, R.K., Shukla, P.K., Bouye, M.,
- Hingaa, S.K., & Mahmoud, A. (2022). Text-Based Emotion Recognition Using Deep Learning Approach. Computational Intelligence and Neuroscience, 2022.
- Bhattacharya, S., Islam, A., & Shahnawaz, S. (2022). TEmoDec: Emotion Detection from Handwritten Text using Agglomerative Clustering. 2022 First International Conference on Artificial Intelligence Trends and Pattern Recognition (ICAITPR), 1-6.
- Bouillon, M., & Anquetil, É. (2015). Handwriting Analysis with Online Fuzzy Models.
- Bride, H., Dong, J., Dong, J.S., & Hóu, Z. (2018). Towards Dependable and Explainable Machine Learning Using Automated Reasoning. IEEE International Conference on Formal Engineering Methods.
- Corke, P. (2021). Images and Image Processing. Robotic Vision.
- Deshpande, S. (2016). GRAPHOLOGY (TO COPY ANY DOCUMENTS IN USER'S HANDWRITING): A REVIEW.
- Deshpande, S. (2016). GRAPHOLOGY (TO COPY ANY DOCUMENTS IN
IN
USER'S HANDWRITING): A REVIEW. International Journal of Advanced Research in Computer Science and Software Engineering, 6(4), 305-310.
- Durga, L., & Deepu, R. (2018). Handwriting Analysis Through Graphology: A Review. 2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI), 1160-1166.
- Durga, L., & Deepu, R. (2018). Handwriting Analysis Through Graphology: A Review. 2018 International Conference on Advances in

Computing, Communications and Informatics (ICACCI), 1160-1166.

<https://doi.org/10.1109/ICACCI.2018.8554463>

- E. Sesa-Nogueras, "Discriminative power of online handwritten words for writer recognition," 2011 Carnahan Conference on Security Technology, Barcelona, Spain, 2011, pp. 1-8, doi: 10.1109/CCST.2011.6095953.
- keywords: {Databases;Handwriting recognition;Error analysis;Accuracy;Text recognition;Biological system modeling;biometrics;writer recognition;handwriting;online words},
- Ekman, P. (1992). Are there basic emotions? *Psychological Review*, 99(3), 550–553. <https://doi.org/10.1037/0033-295X.99.3.550>
- Ekman, P. (1999). Basic emotions. In T. Dalgleish & M. J. Power (Eds.), *Handbook of cognition and emotion* (pp. 45–60). John Wiley & Sons Ltd.
- <https://doi.org/10.1002/0470013494.ch3>
- Esposito, A., Esposito, A. M., Likforman-Sulem, L., Mäizia, P. L., & Zinck, E. (2019). Handwriting and Drawing Features for Detecting Personality Traits. 2019 10th IEEE International Conference on Cognitive Infocommunications (CogInfoCom), 79-84. <https://doi.org/10.1109/CogInfoCom47531.2019.9089985>
- Fairhurst, M., Erbilek, M., & Li, C. (2017). Exploring gender bias in automatic handwriting analysis. *IEEE Transactions on Information Forensics and Security*, 12*(9), 2058-2067.
- Frijda, N. (1969). Recognition of Emotion. *Advances in Experimental Social Psychology*, 4, 167-223.
- Frijda, N. (1969). Recognition of Emotion. *Advances in Experimental Social Psychology*, 4, 167-223. [https://doi.org/10.1016/S0065-2601\(08\)60075-5](https://doi.org/10.1016/S0065-2601(08)60075-5)
- Giezendanner, S., Fisenz, A., Urwyler, P., Walther, S., Perrig, W. J., & Koenig, T. (2018). Impaired concealment of issues in past life: An EEG study

- on how emotional facial expression affects memory retrieval. *Brain and Behavior, 8*(5), e00954.
- Goumiri S, Benboudjema D, Pieczynski W. A new hybrid model of convolutional neural networks and hidden Markov chains for image classification. Neural Comput Appl. 2023 May 31:1-16. doi: 10.1007/s00521023-08644-4. Epub ahead of print. PMID: 37362578; PMCID: PMC10230497.
- Gritčina, L., Križanić, K., & Žaja, R. (2021). Detecting deception from handwriting using machine learning methods. *Expert Systems with Applications, 166,* 114106.
- Guyon, I., et al. (2019). Handwriting recognition competition. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 41*(12), 28422854.
- Hardan, H. (2018). Image Processing. Biomedical Engineering / Biomedizinische Technik.
- Hashemi, S., Vaseghi, B., & Torgheh, F. (2015). Graphology for Farsi Handwriting Using Image Processing Techniques.
- Hernández Millán, Á.R., Mendoza-Moreno, M.A., Portocarrero López, L.M., &
- Castro-Romero, A. (2018). Comparative Study of Machine Learning
- Supervised Techniques for Image Classification Using an Institutional Identification Documents Dataset. 2018 Congreso Internacional de Innovación y Tendencias en Ingeniería (CONIITI), 1-6.
- Hurst, M., Jackson, T.W., & Glencross, M. (2012). Emotion recognition — Theory or practicality. 18th International Conference on Automation and Computing (ICAC), 1-6.
- Joshi, P.M., Ghaskadbi, P., & Tendulkar, S.V. (2018). A Machine Learning Approach to Employability Evaluation Using Handwriting Analysis. Communications in Computer and Information Science.

- Jubair, M.I., & Banik, P. (2013). An Approach to Extract Features from Document Image for Character Recognition. Global journal of computer science and technology, 13.
- K. Nadeem, M. Ahmad and M. Asif Habib, "Emotional States Detection Model from Handwriting by using Machine Learning," 2022 International Conference on Frontiers of Information Technology (FIT), Islamabad, Pakistan, 2022, pp. 284-289, doi: 10.1109/FIT57066.2022.00059.

keywords:

- {Human computer interaction;Emotion recognition;Handwriting recognition;Visualization;Anxiety disorders;Human factors;Machine learning;Emotion recognition;Graphology;Handwriting analysis;Handwriting features;Machine learning},
- Katariya, S., Bhatnagar, R., & Vaidya, T. (2018). Image Processing Based Apporoach Shubham Katariya. International journal of engineering research and technology, 2.
- Kedar, S.V., & Kulkarni, S. (2015). Personality Identification through Handwriting Analysis: A Review.
- Kołakowska, A. (2013). A review of emotion recognition methods based on keystroke dynamics and mouse movements. 2013 6th International Conference on Human System Interactions (HSI), 548-555.
- Kołakowska, A. (2013). A review of emotion recognition methods based on keystroke dynamics and mouse movements. 2013 6th International Conference on Human System Interactions (HSI), 548-555.
<https://doi.org/10.1109/HSI.2013.6577879>
- Kulakov, Y., Tereikovska, L., & Tereikovskiy, I. (2021). Way of application of convolutional neural networks for personality recognition and user emotions by keyboard handwriting. Legal, regulatory and metrological support of information security system in Ukraine.
- L. Likforman-Sulem, A. Esposito, M. Faundez-Zanuy, S. Cléménçon and G. Cordasco, "EMOTHAW: A Novel Database for Emotional State

Recognition From Handwriting and Drawing," in IEEE Transactions on Human-Machine Systems, vol. 47, no. 2, pp. 273-284, April 2017, doi:

- 10.1109/THMS.2016.2635441. keywords: {Databases;Writing;Stress;Feature extraction;Emotion recognition;Atmospheric measurements;Particle measurements;Affective database;anxiety;depression;depression–anxiety– stress scales (DASS);emotional state; handwriting;random forests;stress},
- Le, D., & Provost, E.M. (2013). Emotion recognition from spontaneous speech using Hidden Markov models with deep belief networks. 2013 IEEE Workshop on Automatic Speech Recognition and Understanding, 216-221.
- Lerner, J. S., Li, Y., Valdesolo, P., & Kassam, K. S. (2015). Emotion and decision making. *Annual Review of Psychology, 66,* 799-823.
- Lerner, J. S., Li, Y., Valdesolo, P., & Kassam, K. S. (2015). Emotion and decision making. Annual Review of Psychology, 66, 799-823.
- <https://doi.org/10.1146/annurev-psych-010213-115043>
- Li, Q., Yang, Z., Liu, S., Dai, Z., & Liu, Y. (2015). The study of emotion recognition from physiological signals. 2015 Seventh International Conference on Advanced Computational Intelligence (ICACI), 378-382.
- Lin, Z., & Song, J. (2019). Neural Hidden Markov Model. International Conference on Agents and Artificial Intelligence.
- Lowe, D.G., Bitouk, D., Dhillon, S., Belhumeur, P.N., & Nayar, S.K. (2015). A Survey on Image Processing and Human Action Recognition.
- M. Azmi, S. L. Fathima and M. Wasid, "Unveiling Emotions through Handwriting: A Data Analysis Approach," 2023 12th International Conference on Advanced Computing (ICoAC), Chennai, India, 2023, pp. 1-6, doi: 10.1109/ICoAC59537.2023.10250008. keywords: {Dimensionality reduction;Emotion recognition;Data analysis;Correlation;Predictive models;Real-time systems;Task analysis;EMOTHAW;DASS score;acceleration;strokes;machine learning;feature selection},
- M. S (2022). Mental Health Status Detection through Handwriting Analysis. <https://doi.org/10.22214/ijraset.2022.39897>

- Mantri, R., Meshram, D., Kawale, A., & Jadhav, D. (2022). Handwritten Digit Recognition for Two Digits Using CNN & Image Processing. 2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT), 1-6.
- Mishra, P., Pai, P., Patel, M., & Sonkusare, R. (2020). Extraction of Information from Handwriting using Optical Character recognition and Neural Networks. 2020 4th International Conference on Electronics, Communication and Aerospace Technology (ICECA), 1328-1333.
- Mishra, P., Pai, P., Patel, M., & Sonkusare, R. (2020). Extraction of Information from Handwriting using Optical Character recognition and Neural Networks. 2020 4th International Conference on Electronics, Communication and Aerospace Technology (ICECA), 1328-1333.
- Mohammad Niyaz Khan, S., & Mohd Ghazali, J. (2019). Graphology Analysis for Handwritten Latin and Arabic. Malaysian Journal of Information and Communication Technology (MyJICT).
- Paget, R.A. (1940). Analysis of Handwriting. Nature, 145, 645-645.
- Paluszek, M., & Thomas, S.J. (2019). An Overview of Machine Learning. MATLAB Machine Learning Recipes.
- Premananada, R., Jambukesh, H.J., Shridhar, H., Rajashekar, U., & Harisha,
- K.S. (2020). Design and Implementation of Automated Image Handwriting
- Sentences Recognition using Hybrid Techniques on FPGA. 2020 Third International Conference on Multimedia Processing, Communication & Information Technology (MPCIT), 152-161.
- Pulido-Castro, S., Palacios-Quecan, N., Ballen-Cardenas, M.P., CancinoSuarez, S., Rizo-Arevalo, A., & López, J.M. (2021). Ensemble of Machine Learning Models for an Improved Facial Emotion Recognition. 2021 IEEE URUCON, 512-516.
- Reddy, D.K. (2023). Emotion Recognition through Text, Speech and Image. International Journal for Research in Applied Science and Engineering Technology.

- Reddy, D.K. (2023). Emotion Recognition through Text, Speech and Image.
- International Journal for Research in Applied Science and Engineering Technology. <https://doi.org/10.22214/ijraset.2023.48813>
- Reese, L., & Iberle, K. (2011). Kanban - what is it and why should I care?
- Regli, J., & Nelson, J.D. (2016). Scattering convolutional hidden Markov trees. 2016 IEEE International Conference on Image Processing (ICIP), 1883-1887.
- Robles-Bykbaev, V., López-Gil, J. M., Arnau, D., & Blig, L. (2021). Analysis of handwriting in educational environments to detect emotional states.
- *Journal of Ambient Intelligence and Humanized Computing, 12*(2), 24692483.
- Rocha, A. (2009). Classifiers and machine learning techniques for image processing and computer vision.
- S. V. Kedar, D. S. Bormane, A. Dhadwal, S. Alone and R. Agarwal,
- "Automatic Emotion Recognition through Handwriting Analysis: A Review," 2015 International Conference on Computing Communication Control and Automation, Pune, India, 2015, pp. 811-816, doi: 10.1109/ICCUBEA.2015.162. keywords: {Feature extraction;Emotion recognition;Writing;Handwriting recognition;Databases;Psychology;Image segmentation;Handwriting analysis;Emotion recognition;Emotion outlays;Handwriting features},
- Saltz, J.S., & Heckman, R. (2020). Exploring Which Agile Principles Students Internalize When Using a Kanban Process Methodology. J. Inf. Syst. Educ., 31, 51-60.
- Samsuryadi Samsuryadi, Rudi Kurniawan, F. Mohamad (2021). Automated handwriting analysis based on pattern recognition: a survey. <https://doi.org/10.11591/IJEECS.V22.I1.PP196-206>

- Samsuryadi, S., Kurniawan, R., & Mohamad, F. (2021). Automated handwriting analysis based on pattern recognition: a survey. *Indonesian Journal of Electrical Engineering and Computer Science*, 22(1), 196-206. <https://doi.org/10.11591/ijeecs.v22.i1.pp196-206>
- Sarmiento, C., & Savage, J. (2020). Comparison of Two Objects Classification Techniques using Hidden Markov Models and Convolutional Neural Networks. *Intelligenza Artificiale*, 19, 1222-1254.
- Schröter, A., Mergl, R., Bürger, K., Hampel, H., Möller, H. J., & Hegerl, U. (2003). Kinematic handwriting deficits in Alzheimer's disease. *Journal of Alzheimer's Disease*, 5*(3), 215-228.
- Sen, A., & Shah, H. (2017). Automated handwriting analysis system using principles of graphology and image processing. 2017 International Conference on Innovations in Information, Embedded and Communication Systems (ICIIECS), 1-6.
- Sen, A., Shah, H., Lemos, J., & Bhattacharjee, S. (2018). An Algorithm to Extract Handwriting Feature for Personality Analysis.
- Shalloway, A. (2012). NOLA12-Session1009-02.
- Shammi Akhtar, M. Dipti, Tahasina Afroze Tinni, Pallab Khan, Raihan Kabir, Md. Rashedul Islam (2021). Analysis on Handwriting Using Pen-Tablet for Identification of Person and Handedness. <https://doi.org/10.1109/ICICT4SD50815.2021.9397018>
- Sharma, R., Rajvaidya, H., Pareek, P., & Thakkar, A. (2019). A Comparative Study of Machine Learning Techniques for Emotion Recognition. *Emerging Research in*
- Thakur, M.K., & Goundkar, R.R. (2019). Evaluation of Content Using Machine Learning. *Smart Intelligent Computing and Applications*.
- The, Udeh, C.P., Chen, L., & Wu, M. (2021). Facial Emotion Recognition Using Convolution Neural Networks-Based Deep Learning Model.
- Tilagul, M.A., Asha, G., Dhanushree, S., Harshitha N., A., & Harshitha, S. (2022). OCR on English Handwritten Text. *International Journal of Advanced Research in Science, Communication and Technology*.

- Tso, B., & Olsen, R.C. (2005). Combining spectral and spatial information into hidden Markov models for unsupervised image classification. *International Journal of Remote Sensing*, 26, 2113 - 2133.
- Wagle, K., Rodrigues, S., & Singh, A.D. (2020). Emotion Recognition through Mobile Application.
- Yizhi, C., & Jinchang, L. (2010). Analysis of Handwriting's Characters Based on Method of Functional Data. 2010 International Conference on Computational and Information Sciences, 367-371.
- Zayat, W., & Senvar, O. (2020). Framework Study for Agile Software Development Via Scrum and Kanban. *International Journal of Innovation and Technology Management*, 17, 2030002.
- Zhang, D., Wan, B., & Ming, D. (2015). [Research progress on emotion recognition based on physiological signals]. *Sheng wu yi xue gong cheng xue za zhi* = *Journal of biomedical engineering* = *Shengwu yixue gongchengxue zazhi*, 32 1, 229-34
- Zhang, J., Yin, Z., Chen, P., & Nichele, S. (2020). Emotion recognition using multi-modal data and machine learning techniques: A tutorial and review. *Inf. Fusion*, 59, 103-126.

APPENDICES

APPENDIX A - Informed Consent Form/Assent Form

INFORMED CONSENT

This informed consent form is for 300 minimum right-handed participants (aged 18–25) from a controlled university environment. Participants will provide handwriting samples using standardized tools (0.3mm pen/A4 paper), while self-reporting emotions via questionnaires, who we are invited to participate in this study titled “TITLE OF THE RESEARCH STUDY.”

Name of Researcher(s)/Proponent(s): Cyrel Jane A. Edaño

Name of Organization/Institution: University of San Carlos - Bachelor of Science in Computer Science - DCISM

Title of the Study and Version number: Emotion Recognition through Handwriting Analysis Using a Hybrid Approach 1/22May2025 [e.g. ver. 1/22Feb2024]

This Informed Consent Form has two parts:

- **Information Sheet (to share information about the study with you)**
- **Certificate of Consent (for signatures if you choose to participate)**

PART I: INFORMATION SHEET

Introduction

Hello, I'm Cyrel Jane Edaño, a Computer Science student at the University of San Carlos. I'm conducting research on "Emotion Recognition through Handwriting Analysis Using a Hybrid Approach" and invite you to participate. You don't need to decide today, feel free to discuss this with anyone you trust and take time to reflect. If any terms or concepts are unclear, I'll explain them as we proceed, and you may ask questions at any time.

This consent form may contain words that you may not understand. Please ask me to stop as we go through the information and I will take time to explain. If you have questions later, you can ask them of me or of another researcher.)

Purpose of the Research

We all express emotions in unique ways even through our handwriting. This study explores whether changes in how we write (like letter shapes, spacing, or pressure) can reflect feelings of happiness or sadness. By participating, you'll

help us understand these connections better. Your contribution could one day support tools for educators or counselors to recognize emotional needs earlier but this study focuses only on discovering handwriting patterns, not providing personal analysis.

Type of Research Intervention

This study involves a single, in-person session where you'll complete handwritten tasks using materials we provide (paper and pen). You'll be asked to:

- Fill out a short questionnaire about your current emotions,
- Copy specific words and sentences in cursive handwriting, and
- Complete simple drawing exercises (like sketching shapes or a clock).

The session will take 30–45 minutes and will be conducted privately. Your handwriting will be scanned afterward to analyze its patterns no medical procedures, physical tests, or group activities are involved.

Participant Selection

You're being invited because your profile as a right-handed young adult (18–25 years) with cursive handwriting skills matches the focus of this study. We're seeking participants like you to help uncover how handwriting patterns may relate to emotions across this specific age group. Your participation will contribute to building tools that could someday support student well-being in educational settings.

Voluntary Participation

Joining this study is entirely your choice you may participate or decline without any penalty or effect on your academic standing, university services, or future opportunities. If you choose to participate, you can withdraw at any time, for any reason, without explanation. Even if you complete the handwriting tasks, you may request your data be removed before analysis. The research may end early if technical issues arise or if ethical concerns emerge.

Procedures

We are inviting you to take part in a study that explores how handwriting can reflect emotional states, specifically happiness and sadness. If you agree, we will first explain the study and provide you with an Informed Consent Form to sign.

You will then answer a short questionnaire about how you feel right now. You may skip any question you are not comfortable with.

Next, you will be asked to copy some words and write sentences on a sheet of paper using a black pen. Your handwriting will be scanned and analyzed using a computer to detect emotional patterns based on writing style not content.

Your name will not appear on any documents. You will be identified only by a code. Only the researcher, Cyrel Jane A. Edaño, will have access to your data. All information is confidential and will be securely stored, then deleted after two years.

Duration

The research will take place over a period of four months. However, each participant's involvement will only take about 10 to 15 minutes in total. During that time, you will be asked to answer a short questionnaire and complete a handwriting task.

There are no follow-up sessions required after your participation. All collected data both printed and digital will be securely deleted two years after the study is completed.

Risks (Including possible discrimination, burden, discomforts, and inconveniences)

There are five acknowledged risks generally associated with participation in research studies physical, psychological, social, economic, and legal. However, the researcher foresees minimal risk for those who choose to participate in this study.

There are no physical risks involved. Some participants may feel mild discomfort when reflecting on their emotional state or while completing the writing task. If at any point you feel uneasy, you are free to skip any question or withdraw from the study entirely without giving a reason.

There are no expected risks of discrimination, social judgment, or legal issues, as your name will not appear on any documents and your responses will remain confidential. The researcher will ensure that your data is securely handled and disposed of after three years. If any discomfort continues after participation, you may contact the faculty adviser for support (contact information is provided below).

Foreseeable circumstances and reasons under which participation in the study may be terminated

Your participation in this study may be terminated if any of the following circumstances occur:

- You choose to withdraw from the study at any time. Participation is voluntary, and you are free to stop without providing any reason.
- You do not complete the questionnaire or handwriting task as required for the study.
- The researcher determines that continuing your participation may cause you discomfort or harm.
- Technical issues occur (e.g., poor quality handwriting scans) that prevent your data from being properly analyzed.

In any case, your data will be excluded from the final analysis, and all associated records will be securely deleted.

Benefits

There will be no direct benefit to you for participating in this study. However, your participation will help us develop a system that can recognize emotional states through handwriting. This could benefit the academic community, contribute to mental health research, and support future innovations in education, psychology, and human-computer interaction.

Reimbursement as a result of participation in the research

You will not be provided with any incentive to take part in the research. However, you may receive a small token of appreciation or reimbursement (e.g., for transportation or time) if applicable, depending on the availability of research funds. This is to compensate for any minor inconvenience your participation may cause.

Compensation and/or medical treatments in case of injury

This study does not involve any physical procedures or medical interventions, and the risk of injury is extremely low. Therefore, no compensation or medical treatment will be provided. However, if you experience any discomfort or emotional distress during the study, you may stop at any time and seek support. You may also contact the researcher or faculty adviser for assistance.

Confidentiality

The information that we collect from this research project will be kept strictly confidential. Your name will not appear on any of the materials you complete. Instead, a code number will be assigned to your responses and handwriting

samples. Only the researcher, Cyrel Jane A. Edaño, will know which code corresponds to which participant.

All printed documents such as consent forms and questionnaires which will be stored in a locked cabinet, while all digital files such as scanned handwriting images and analysis data will be stored in a password-protected computer.

Your data will not be shared with anyone outside the research team, and will only be used for academic purposes. All files and documents will be permanently deleted or shredded three years after the completion of the study.

This study does not involve highly sensitive topics, and efforts will be made to ensure that your identity and responses remain anonymous. However, if you choose to talk about your participation with others, people may be able to guess you were involved. You are not required to discuss your participation with anyone.

Sharing the Result

Nothing you share in the questionnaire or handwriting task will be shared with anyone outside the research team, and nothing will be linked to your name. After the study is completed and the results have been analyzed, a summary of the findings will be made available to all participants upon request. The researcher may also present the results at academic conferences or publish them in research journals, but no personal or identifying information will ever be included.

Right to Refuse or Withdraw

Your participation in this study is entirely voluntary. You do not have to take part if you do not wish to, and choosing not to participate will not affect you in any way. You may also withdraw from the study at any time, even after you have agreed to participate, without giving any reason. If you choose to stop, any information you have provided will not be used and will be securely deleted.

Who to Contact

If you have any questions or concerns about this research, you may contact the primary researcher:

Cyrel Jane A. Edaño

Email: 20101954@usc.edu.ph

You may also contact the faculty adviser:

Christian V. Maderazo, MEng

Email: cvmaderazo@usc.edu.ph

For concerns regarding your rights as a participant or ethical issues in this study, you may contact:

University of San Carlos – Research Ethics Committee (USC-REC)
Email: rec@usc.edu.ph
Phone: +63 (32) 4012300 local 204

If you wish to ask questions later or in the event of any research-related injury, you may contact any of the following:

Name of Researcher(s) : Cyrel Jane A. Edaña
Course : Bachelor of Science in Computer Science
University of San Carlos
Email: 20101954@usc.edu.ph

This study has undergone ethics review by the University of San Carlos Research Ethics Committee and may be reached regarding rights of the study participants, including grievances and complaints through the contact information below:

Rochelle Marie E. Leros, M.A., M.E.N.R.M.

Chair, USC Research Ethics Committee
University of San Carlos
Nasipit, Talamban, Cebu City
Email: rec@usc.edu.ph
Landline: (032) 4012300 local 204

PART II: CERTIFICATE OF CONSENT

I have been invited to participate in research about recognizing emotional states, specifically happiness and sadness through handwriting analysis using image processing techniques.

I have read (or have had read to me) the information provided about this study. I have had the opportunity to ask questions and any questions I have asked have been answered to my satisfaction. I know that my participation is voluntary and that I can withdraw at any time without penalty or loss of benefits.

I have read the foregoing information, or it has been read to me. I have had the opportunity to ask questions about it and any questions I have been asked have been answered to my satisfaction. I consent voluntarily to be a participant in this study

Respondent No. _____

Print Name of Participant _____

Signature of Participant _____

Phone Number _____

Email: _____

Postal Address: _____

Date _____

Day/month/year

If illiterate

I have witnessed the accurate reading of the consent form to the potential participant, and the individual has had the opportunity to ask questions. I confirm that the individual has given consent freely.

Print name of witness _____
participant

Thumb print of

Signature of witness _____

Date _____

Day/month/year

Statement by the researcher/person taking consent

I have accurately read out the information sheet to the potential participant, and to the best of my ability made sure that the participant understands that the following will be done:

1. Complete a self-report questionnaire assessing their current emotional state.
2. Perform drawing tasks, including copying geometric shapes (pentagons, house), drawing loops with both hands, sketching a clock, and creating a self-portrait.
3. Complete handwriting tasks:
 - Write specific words in CAPITAL LETTERS using left and right hands.
 - Transcribe sentences in cursive script.

I confirm that the participant was given an opportunity to ask questions about the study, and all the questions asked by the participant have been answered correctly and to the best of my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily.

A copy of this ICF has been provided to the participant.

Print Name of Researcher/person taking the consent _____

Signature of Researcher /person taking the consent _____

Date _____

Day/month/year

APPENDIX B - Questionnaire/Research instrument

AGE:	GENDER:
------	---------

Assessment Instructions
This assessment helps us understand your current emotional state and cognitive functions through simple exercises. Please complete all sections to the best of your ability.

Part 1: Self-Report Questions

- Read each statement and place a **Checkmark (✓)** next to those that apply to you
- Be honest – there are no right or wrong answers
- Take your time to consider each statement carefully

Part 2: Drawing Tasks

1. Draw three overlapping circles
2. Connect the Dots
3. Draw a Person in the Rain
4. Draw a House and a Tree

Part 3: Writing Tasks

1. Write the following words:
 - With your **LEFT** hand:
 - With your **RIGHT** hand:
 - Rewrite all words in **UPPERCASE** letters
2. Write the following sentences in **CURSIVE**:

Important Notes:

- Please complete all sections in one sitting
- Take breaks between sections if needed
- There is no time limit – work at your own comfortable pace
- Please do not erase any of your work.

Cyrel Jane A. Edaña
Researcher

Christian V. Maderazo, M.Eng.
Adviser

Caira Wynn Louise B. Syntig, RPm.
Psychometrician

Hazel T. Suquib, RPm.
Psychometrician

Peter John Toreta, RPm.
Psychometrician

Alden M. Arda, RPm.
Psychometrician

Part 1: The SELF-REPORT

Below are 24 items. Please indicate how much you agree or disagree with each by put a **CHECKMARK✓** in the blank after each statement, according to the following scale:

1 = Strongly Disagree	2 = Disagree	3 = Neutral	4 = Agree	5 = Strongly Agree
-----------------------	--------------	-------------	-----------	--------------------

Read each statement carefully. Some are phrased positively, others negatively. Answer quickly there are no right/wrong answers. Trust your first instinct. If unsure, select what applies most of the time.

	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>
I feel emotionally drained.					
I feel enthusiastic about my activities.					
I find it difficult to look forward to things.					
I wake up feeling positive about the day ahead.					
I feel a sense of emptiness.					
I laugh easily.					
I experience a heaviness in my chest.					
I feel a general sense of well-being.					
I have less interest in activities I usually enjoy.					
I feel at peace with myself.					
I feel less optimistic than usual.					
I experience a sense of contentment with my life.					
I feel a sense of purpose in my daily activities.					
My thoughts tend to be more negative than positive.					
I feel like crying.					
I feel grateful for what I have.					
I feel disconnected from others.					
My energy levels are lower than usual.					
I am satisfied with how things are going in my life.					
I find it easy to enjoy the present moment.					
I feel optimistic about the future.					
I feel a general sense of loneliness.					
I feel cheerful most of the time.					
I feel less enthusiastic about activities I once enjoyed.					

Part 2: The Drawing Exercise

Draw a Three Overlapping Circles	Connect the Dots
	
Draw a Person in the Rain	Draw a House and a Tree

Part 3: The Writing Exercise

Write the following words			
	Write here using Left Hand	Write here using Right Hand	Write here in Uppercase
Content			
Melancholic			
Optimistic			
Disconnected			
Vibrant			

Write the following sentences in CURSIVE letters*Crazy people are seeking for purple flowers, drinking water and dreaming time**Happiness is the meaning and the purpose of life, the whole aim and end of human existence.**Life is like a box of chocolates.**What does not kill me makes me stronger.**Happiness is not doing what you want, but wanting what you do.*

APPENDIX C – Transmittal Letter



May 13, 2025

Caira Wynn Louise B. Yntig, Rpm.
Psychometrician

Dear Miss Yntig,

Greetings!

May I request your expertise to validate the psychological assessment protocols for my thesis, "INSIDE-OUT: EMOTION RECOGNITION THROUGH HANDWRITING ANALYSIS USING A HYBRID APPROACH". The above project is a requirement for the completion of Computer Science Thesis 1 in the Department of Computer, Information Sciences, and Mathematics. Your guidance in ensuring ethical compliance, accuracy of emotion annotations (happiness/sadness), and alignment with psychological theories would significantly enhance the study's reliability for potential applications in education and mental health. I am happy to provide clarifications or adjustments as needed. I am confident that the result of the study would be useful and helpful in the betterment of the community.

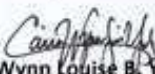
Thank you very much. I am looking for your positive response.
Respectfully yours,


Cyrel Jane A. Edaño
CS Thesis 1 Student

Endorsed by:


Christian V. Maderazo, MEng
Thesis Adviser

Approved by:


Caira Wynn Louise B. Yntig, Rpm.
Psychometrician



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DEPARTMENT OF COMPUTER, INFORMATION SCIENCES AND MATHEMATICS

May 13, 2025

Hazel T. Suquib, RPs.
Psychometrician

Dear Miss Suquib,

Greetings!

May I request your expertise to validate the psychological assessment protocols for my thesis, "INSIDE-OUT: EMOTION RECOGNITION THROUGH HANDWRITING ANALYSIS USING A HYBRID APPROACH". The above project is a requirement for the completion of Computer Science Thesis 1 in the Department of Computer, Information Sciences, and Mathematics. Your guidance in ensuring ethical compliance, accuracy of emotion annotations (happiness/sadness), and alignment with psychological theories would significantly enhance the study's reliability for potential applications in education and mental health. I am happy to provide clarifications or adjustments as needed. I am confident that the result of the study would be useful and helpful in the betterment of the community.

Thank you very much. I am looking for your positive response.
Respectfully yours,

Cyrel Jane A. Edaño
CS Thesis 1 Student

Endorsed by:

Christian V. Madelazo, MEng
Thesis Adviser

Approved by:

Hazel T. Suquib, RPs.
Psychometrician



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DEPARTMENT OF COMPUTER, INFORMATION SCIENCES AND MATHEMATICS

May 13, 2025

Alden M. Arda, RPs.
Psychometrician

Dear Mr. Arda,

Greetings!

May I request your expertise to validate the psychological assessment protocols for my thesis, "INSIDE-OUT: EMOTION RECOGNITION THROUGH HANDWRITING ANALYSIS USING A HYBRID APPROACH". The above project is a requirement for the completion of Computer Science Thesis 1 in the Department of Computer, Information Sciences, and Mathematics. Your guidance in ensuring ethical compliance, accuracy of emotion annotations (happiness/sadness), and alignment with psychological theories would significantly enhance the study's reliability for potential applications in education and mental health. I am happy to provide clarifications or adjustments as needed. I am confident that the result of the study would be useful and helpful in the betterment of the community.

Thank you very much. I am looking for your positive response.
Respectfully yours,

Cyrel Jade A. Edaña
CS Thesis 1 Student

Endorsed by:

Christian V. Maderazo, MEng
Thesis Adviser

Approved by:

Alden M. Arda, RPs.
Psychometrician



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SCIENTIA ET HUMANITAS



DEPARTMENT OF COMPUTER, INFORMATION SCIENCES AND MATHEMATICS

May 13, 2025

Peter John Toreta, RPM.
Psychometrician

Dear Mr. Toreta,

Greetings!

May I request your expertise to validate the psychological assessment protocols for my thesis, "INSIDE-OUT: EMOTION RECOGNITION THROUGH HANDWRITING ANALYSIS USING A HYBRID APPROACH". The above project is a requirement for the completion of Computer Science Thesis 1 in the Department of Computer, Information Sciences, and Mathematics. Your guidance in ensuring ethical compliance, accuracy of emotion annotations (happiness/sadness), and alignment with psychological theories would significantly enhance the study's reliability for potential applications in education and mental health. I am happy to provide clarifications or adjustments as needed. I am confident that the result of the study would be useful and helpful in the betterment of the community.

Thank you very much. I am looking for your positive response.
Respectfully yours,


Cyrel Jane A. Edaño
CS Thesis 1 Student

Endorsed by:


Christian M. Magarazo, MEng
Thesis Adviser

Approved by:


Peter John Toreta, RPM.
Psychometrician

APPENDIX D – Approval Sheet



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Date: May 22, 2025

APPROVAL SHEET

This thesis proposal entitled: **INSIDE-OUT: Emotion Recognition through Handwriting Analysis using Hybrid Approach**

submitted by: **CYREL JANE A. EDAÑO** for the degree of: **BACHELOR OF SCIENCE IN COMPUTER SCIENCE** has been approved by the Committee assigned to the students.

Handwritten signature of Christian V. Maderazo in black ink.

CHRISTIAN V. MADERAZO, M.Eng.
Adviser

Handwritten signature of Archival J. Sebial in black ink.

Archival J. Sebial, DIT
Chair

Handwritten signature of Khent L. Dela Paz in black ink.

KHENT L. DELA PAZ
Member

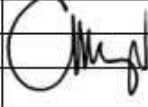
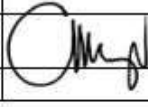
Handwritten signature of Engr. Patrick Roy M. Elalto in black ink.

Engr. PATRICK ROY M. ELALTO
Member

Handwritten signature of Angie M. Ceniza-Camillo in black ink.

ANGIE M. CENIZA-CAMILLO, PhD
Thesis Proposal Coordinator

THESIS COMMITTEE RECOMMENDATIONS
COMPLIANCE FORM
(Proposal Hearing)

SUGGESTIONS	NOT APPLICABLE	COMPLIED WITH	
		YES Page No.	NO Reason/s
1. Title		<u>i</u>	
2. Abstract		<u>ii</u>	
3. Rationale of the Study		<u>2</u>	
4. Statement of the Problem		<u>3</u>	
4.1 General Objective		<u>3</u>	
4.2 Specific Objectives		<u>4</u>	
5. Significance of the Study		<u>4</u>	
6. Scope and Limitations		<u>5</u>	
7. Review of Related Literature		<u>6</u>	
8. Technical Background		<u>15</u>	
9. Design and Methodology		<u>19</u>	
9.1 Conceptual Framework		<u>20</u>	
9.2 Analysis and Design		<u>21</u>	
9.3 Development Model		<u>22</u>	
9.4 Development Approaches		<u>23</u>	
9.5 Software Development Tools		<u>25</u>	
9.6 Project Management		<u>26</u>	
9.6.1 Schedule and Timeline		<u>26</u>	
9.6.2 Responsibilities		<u>26</u>	
9.6.3 Budget and Cost Management		<u>28</u>	
9.7 Verification, Validation and Testing		<u>29</u>	
10. Bibliography		<u>32</u>	
11. Appendices		<u>42</u>	
12. Curriculum Vitae		<u>50</u>	

Submitted by:


CYREL JANE A. EDAÑO
Name and Signature of Candidate


CHRISTIAN V. MADRAZO, M.Eng.
Name and Signature of Adviser



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GENERAL INSTRUCTIONS: Please follow the steps below in submitting your REVISED THESIS PROPOSAL DOCUMENT

1. Kindly fill up the entries of the two forms: APPROVAL SHEET and THESIS COMMITTEE RECOMMENDATION COMPLIANCE FORM.
2. Attached the APPROVAL SHEET and THESIS COMMITTEE RECOMMENDATION COMPLIANCE FORM in your REVISED THESIS DOCUMENT.
3. Fill-out the THESIS COMMITTEE RECOMMENDATION COMPLIANCE FORM for reference in checking the revised thesis document and make sure to affix your signatures.
4. Let your THESIS ADVISER do the following:
 - a. Review your REVISED THESIS PROPOSAL DOCUMENT
 - b. Sign the THESIS COMMITTEE RECOMMENDATION COMPLIANCE FORM
5. Route your THESIS PROPOSAL DOCUMENT to the PANELIST
 - a. First to the MEMBERS of the PANEL
 - b. Lastly to the CHAIR of the PANEL
6. Have your APPROVAL SHEET signed by the Thesis Proposal Coordinator
7. And, SUBMIT your signed approved thesis document to your research coordinator.

APPENDIX E - CV of researcher

CURRICULUM VITAE

CONTACT INFORMATION

Full Name: Cyrel Jane A. Edaño

Current Address: Jagna, Bohol, Philippines, 6308

Telephone number: 09668011383

Email address: edano.cyreljane@gmail.com



PERSONAL INFORMATION

Date of Birth: February 20, 2001

Age: 24

Citizenship/Nationality: Filipino

Gender: Female

Marital Status: Single

Number of Children & Age: None

Language proficiency: English, Filipino, Cebuano, Spanish A1

Software skills: Java, Javascript, NodeJS, Python, LaTeX, Php, TypeScript, Jupyter, R, Sql, Flutter, Figma, AWS, Google Cloud, Azure, Git, React, Django, Numpy, Scikit-Learn

EDUCATIONAL BACKGROUND

1. Education Level

- University of San Carlos – Talamban Campus
Bachelor of Science in Computer Science
Sitio Nasipit, Brgy. Cebu City, Cebu, 600
Tertiary Level(2020-present)
- Central Visayan Institute Foundation, Inc.
Information and Communications Technology
Brgy. Looc, Jagna, Bohol, 6308
Senior Secondary Level (2018-2020)
- Central Visayan Institute Foundation, Inc.
Brgy. Looc, Jagna, Bohol, 6308

Junior Secondary Level (2014-2018)

- Child and Arts Academy, Inc.
Brgy. Canjulao, Jagna, Bohol, 6308
Elementary Level (2008-2014)

2. Certifications & Accreditations

Google UX Design Professional Certificate

Jan 2024

CyberOps Associate

Dec 2023

Introduction of Data Science

Sept 2023

Introduction of Cybersecurity

Oct 2022

AI Fundamentals with IBM SkillsBuild

June 2025

3. Trainings & Seminars

SEASAT Youth Camp 2023

July 2023

University Research Conferences 2023

April 2023

PSITE Central Visayas Region ICT Student Congress - Cebu

March 2023

23rd Philippine Computing Science Congress

March 2023

STRENGTHS/TRAITS & SKILLS

1. Leadership skills
2. Can work independently with minimal supervision and work with a team
3. Hands-on experience

4. Able to work within tight deadlines

CAREER OBJECTIVES

Further developed career path and skills within data science and cloud computing.

I certify that the information provided in this CV is true and correct to the best of my knowledge.



CYREL JANE A. EDAÑO

SIGNATURE

(JULY 01, 2025)

APPENDIX F - CV of Adviser

1. Name:				
Maderazo <i>Family Name</i>		Christian <i>First Name</i>		Valencia <i>Middle Name</i>
2. Gender: ☐ Male ☐ Female	3. Age: 52	4. Birthplace: Sagay City, Negros Occidental	5. Date of Birth: December 25, 1972	6. Citizenship: Filipino
7. Civil Status: Married	8. Department: Department of Computer and Information Sciences	9. Position: Department Head		10. Academic Rank: Asst Professor 4
11. Home Address: 1204 Talamban, Cebu City		12. Telephone Number: 09173811746 13. E-Mail Address: cvmaderazo@usc.edu.ph		
14. Schedule of Research: 8-10 AM Monday to Saturday				

15. Teaching Experience: (No. of Years) In Other Institutions 0 At USC 29	16. Date Employed at USC: June 1996	17. Status of Employment: Permanent
18. Previous Positions held at USC		
Date	Position	
	Digital Microprocessor Lab Head	
	CEAC – Engineering Lab Head	
	Computer Networking Lab Head	
	Software Engineering Lab Head	

19. Work Experience:		
Date	Employer's Name and Address	Position
June 1996	University of San Carlos – Cebu	Faculty
20. Educational Background:		
Degree Earned	Institution	Date Obtained
Elementary:	Eusebio Lopez Memorial School	1986
Secondary:	Holy Trinity Academy	1990
Bachelor's:	University of San Carlos	1996
Master's:	University of San Carlos	2008
Title of Thesis: Automated Examination Generator and Checker		
Thesis Adviser: Dr. Rosana Ferolin		
Doctoral:	Cebu Technological University	Candidate

Title of Dissertation:							
Dissertation Adviser:							
Post Doctoral Studies:							
Special Training/Course Attended							
Java Programming Certification Training							
Licensure Examination Passed							
21. Research Projects Conducted:							
Title		Research Involvement (Please use code*)		Funding Agency		Inclusive Period	
<u>Integrated School Management Information System</u>		<u>Developer</u>		<u>USC</u>		2011-2014	
<u>LIDAR</u>		<u>Research Assistant</u>		<u>DOST</u>		<u>2016-2017</u>	
22. Publications							
Authors name (if w/co-authors, arrange according to sequence as it appears in the article)	Year Published	Title of the article	Title of Journal (Please do not abbreviate)	Volume No.	Issue No.	Inclusive Pages	Type of Journal (Use code*)
Angie Ceniza, Christian Maderazo, Mary Jane Sabellano	2015	Perception Analysis for University of San Carlos (USC) as an Educational Institution using Web Mining and Multinomial Naïve Bayes Algorithm	International Journal of Engineering Research & Technology				3

Glady Mae Payusan, Christian Maderazo, Angie Ceniza	2016	A Decision Support System in Facilitating Properties for Rent using Analytic Hierarchy Process and Goal Programming	2016 International Conference on Advanced Computing, Communication and Information Sciences			79-80	3
Ken Gorro, Mary Jane Sabellanon, Christian Maderazo, Angie Ceniza, Kim Gorro		Exploring Facebook for Sharing Crime Experiences Using Selenium and Support Vector Machine	Association for Computing Machinery			218-222	3
Patrick Dave P. Woogue, Gabriel Andrew A. Pineda, and Christian V. Maderazo		Automatic Web Page Categorization Using Machine Learning and Educational-Based Corpus	International Journal of Computer Theory and Engineering	9	6	427-432	3
Charmaine Llorca, May Elsbeth Yares, Christian Maderazo		International Conference on Technological Challenges for Better World 2018	International Conference on Technological Challenges for Better World 2018				3

(Please use a separate sheet if necessary)

* 1 – Local/Institution Publication, 2 – National Publication, 3 – International Publication, 4 – Others
(Please specify)

23. Current Research Projects:

Title	Target Date of Completion
e-Ligtas	January 2019

(Please use a separate sheet if necessary)

* 1 - Primary Investigator, 2 - Co-Investigator, 3 - Research Assistant, 4 - Data Field Supervisor, 5 - Interviewer, 6 - Encoder, 7 - Data Programmer, 8 - Data Analyst, 9 - Others (Please specify.)

24. Honors and Awards Received:

25. Professional Affiliations:

Philippine Society of Information Technology Educators

Cebu foundation for IT /CIB.O

26. Research-Related

Date	Place	Purpose

I certify that all information contained herein are true and correct.


Christian V. Madarazo
Date: July 1, 2025